UNIT 5 – I/O MANAGEMENT & FILE SYSTEM (EXPLANATORY NOTES)

◇ PART A: I/O MANAGEMENT

◇ 1. What is I/O? (Basic Idea)

In any computer system, every task follows 3 steps:

- Input (data comes in)
- Processing (CPU works)
- Output (result goes out)

Example:

- You type code using keyboard → Input
- CPU executes it → Processing
- Output appears on screen → Output

Problem:

- CPU is **very fast**
- Devices (keyboard, disk, printer) are **slow**

👉 So OS must manage communication between them
👉 This is called **I/O Management**

◇ 2. Types of I/O Devices

Devices are mainly of two types:

◇ 1. Block Devices

- Data is transferred in blocks
- Example: Hard Disk, SSD

👉 When you open a file, data comes in chunks (blocks)

◇ 2. Character Devices

- Data is transferred character by character
- Example: Keyboard, Mouse

👉 When you type, input comes one character at a time

◇ 3. I/O Subsystem (Important Concept)

CPU cannot directly control all devices.
So OS creates a layer called **I/O Subsystem**.

Main Components:

◇ ***Device Driver***

- Software that connects OS with hardware

👉 Example:

- Printer driver in Windows
- GPU driver in Linux

👉 Without driver → device will not work

◇ ***Interrupt Handling***

- Device sends signal when task is complete

👉 Example:
Download finished → notification appears

◇ 4. I/O Buffering (Very Important)

Now understand this clearly:
👉 CPU is fast
👉 Devices are slow

If CPU waits → performance drops

Solution: Buffer

Buffer = temporary memory

Real Example:

When watching YouTube:

- Video loads first (buffer)
- Then plays smoothly

Types of Buffering:

◇ ***Single Buffer***

- One buffer used
- Simple but slow

◇ ***Double Buffer***

- Two buffers
- One used by CPU, one by device

👉 Faster system

◇ ***Circular Buffer***

- Multiple buffers in loop

👉 Used in streaming, networking

◇ 5. Disk Storage (How data is stored)

Disk is used for permanent storage.

Structure:

- Track → circular path
- Sector → small part of track
- Cylinder → group of tracks

👉 Like rings inside a CD

◇ 6. Disk Scheduling (Very Important 🔥)

Many processes request disk access at the same time.

👉 Disk can handle only one request at a time

👉 OS must decide order

Example:

Requests = 98, 183, 37, 122

Which one first?

Algorithms:

◇ FCFS (First Come First Serve)

- Serve in order

👉 Simple but slow

◇ SSTF (Shortest Seek Time First)

- Serve nearest request

👉 Faster

🔹 SCAN (Elevator Algorithm)

- Disk head moves in one direction
- Serves all requests on the way

👉 Like lift

🔹 C-SCAN

- Moves in one direction only
- Jumps back to start

👉 More fair

🔹 LOOK / C-LOOK

- Similar to SCAN but more optimized

Real OS:
Windows and Linux use advanced versions of these

🔹 7. RAID (Important for exam 🔥)

RAID = Redundant Array of Independent Disks
👉 Uses multiple disks together

Why RAID?

- Increase speed
- Improve data safety

Types:

🔹 ***RAID 0***

- Fast
- No backup

🔹 ***RAID 1***

- Mirror copy
- Safe

🔹 ***RAID 5***

- Balance of speed and safety

Real Example:
Used in servers and data centers

◇ PART B: FILE SYSTEM

◇ 8. What is a File?

File = collection of data stored on disk

Examples:

- file.txt
- image.jpg
- program.exe

File Attributes:

- Name
- Size
- Type
- Location
- Permissions

◇ 9. File Organization & Access Methods

OS must know:

👉 Where file is stored

👉 How to access it

Access Methods:

◇ Sequential Access

- Read one by one

👉 Like reading a book

◇ Direct Access

- Jump to any position

👉 Like skipping video

◇ Indexed Access

- Use index to find data

👉 Like book index

◇ 10. File Directories

Directory = folder

Used to organize files

Types:

◇ Single Level

- All files in one folder

👉 messy

◇ Two Level

- Separate folder per user

◇ Tree Structure (Important)

- Folder inside folder

👉 Used in:

- Windows
- Linux
- macOS

◇ 11. File Sharing

Multiple users can use same file

Permissions:

- Read
- Write
- Execute

Example:

Linux permissions:

- r → read
- w → write
- x → execute

◇ 12. File System Implementation

How files are stored on disk:

◇ Contiguous Allocation

- Continuous blocks

👉 Fast but fragmentation

◇ Linked Allocation

- Blocks linked

👉 Flexible but slow

◇ Indexed Allocation

- Index table stores addresses

👉 Best method

◇ 13. File Protection & Security

Protect data from misuse

Methods:

◇ ***Access Control***

Who can use file

◇ ***Permissions***

Read / Write / Execute

◇ ***Encryption***

Convert data into secret form

◇ ***Backup***

Keep copy of data